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PALAEOMAGNETIC STUDY OF LATE PRECAMBRIAN ROCKS OF THE MIDLAND CRATON OF ENGLAND AND WALES

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The Precambrian basement of the British region south of the Caledonian orogenic belt is only observed in a few small inliers; this paper reports a detailed palaeomagnetic study of four of these inliers. The Stanner–Hanter amphibolitised gabbro-dolerite complex of uncertain age yields a mean direction of magnetisation $D = 282°, I = 51°$ (15 sites, $\alpha_{95} = 11.4°$) after AF and thermal cleaning. Uriconian lavas and tuffs (~700–600 m.y.) of the Pontesford and Wrekin inliers require both thermal and AF cleaning for complete analysis of NRM. The former region (Western Uriconian) yields a mean of $D = 136°, I = -25°$ (6 sites, $\alpha_{95} = 15.3°$) and the latter region (Eastern Uriconian) a mean of $D = 78°, I = 17°$ (9 sites, $\alpha_{95} = 12.8°$); the Eastern Uriconian shows a marked improvement in precision after a two-stage fold test, and the palaeomagnetic data suggest that some apparent polar movement took place between eruption of the two sequences. The Uriconian rocks in both areas were intruded by dolerites which yield a mean direction of magnetisation $D = 72°, I = 54°$ (11 sites, $\alpha_{95} = 13.2°$).

The collective data give palaeomagnetic poles related to Upper Proterozoic metamorphic episodes (Stanner–Hanter Complex and Rushton Schist) which are in close agreement with earlier studies of the Malvernian metamorphic rocks, and to the late Precambrian Uriconian volcanic/hypabyssal igneous episode. All of these magnetisations are probably confined to the interval 700–600 m.y., and are indicative of appreciable polar movement during this interval. The palaeomagnetic poles define an apparent polar wander path for this crustal block between Late Precambrian and Lower Cambrian times and show that cratonic Britain south of the Caledonian suture is unrelated to the Baltic Shield.

1. Introduction

Late Precambrian–Cambrian times were characterised by much higher rates of apparent polar wander (a.p.w.) movements than any later time (McElhinny et al., 1974; McElhinny and Embleton, 1976; Piper, 1976a). Identification of these movements is based on data from the integral supercontinent of Gondwanaland, and such is the paucity of palaeomagnetic data from elsewhere that comparisons with contemporaneous data have been considered in the context of the subduction of enormous oceans (Lomax and Briden, 1977) or of large movement relative to the poles of a loosely-coupled grouping of continents (Piper, 1976a). This late Precambrian to Cambrian interval is a particularly important one in the tectonic evolution of the crust because it defines a time of transition: ensialic mobile belts were developing in the Pan African domain (Shackleton, 1973; Piper et al., 1973) with characteristics most typical of earlier Proterozoic events, while the Caledonian belt was developing in association with obduction and subduction of ocean crust (Bird et al., 1971).

To determine whether or not these a.p.w. movements are large and similar with respect to all continents it will be necessary to obtain a detailed time sequence of palaeomagnetic poles from each cratonic unit. In practice, this objective is limited by the variety of rock available for study. Cratonic Britain south of the Caledonian suture, the subject of this study, represents a particular problem firstly because the Precambrian basement is brought to the surface only

within a number of small inliers (Dunning, 1975), and secondly because the Precambrian rocks are in most cases both strongly folded, and faulted against younger rocks. This paper reports a detailed palaeomagnetic study of several of these inliers, and, although the quality of the final results is limited by tectonic considerations, the new data complement some earlier work (Brooks, 1963; Lomax and Briden, 1977) to define an a.p.w. path for Britain south of the Caledonian suture in Late Precambrian times (~700–570 m.y.); they also permit some preliminary comparisons with palaeomagnetic results from the Baltic Shield.

2. Geology

The inlier comprising the Stanner, Hanter and Worsell Hills of Radnorshire (Holgate and Knight-Hallowes, 1941; Kirk, 1951) is composed of an amphibolitised igneous complex. Early calc-alkaline dolerites were intruded and thermally metamorphosed by gabbros prior to widespread amphibolitisation which produced albite, epidote and hornblende, although localised pyroxene remains in both facies. The complex was veined by aplites, acid porphyries and granophyre with affinities with late Precambrian Uriconian rocks elsewhere in the Welsh Borderlands (Holgate and Knight-Hallowes, 1941; Wright, 1969; Baker, 1971). It is intensely cut by faults with several trends, and the present outcrop was fractured in the vicinity of major NE–SW faults which bring the complex into juxtaposition with younger rocks. The sequence of metamorphic events resembles that exhibited by the Malvernian metamorphic assemblage of the Malvern Hills 60 km to the east, and there are some qualitative grounds for correlating the two areas, although the only radiometric age for the Stanner and Hanter complex is a K–Ar Caledonian overprint date of 418 ± 29 m.y. (Fitch et al., 1969). Pebbles from the complex are found in nearby Longmyndian sediments (Holgate and Knight-Hallowes, 1941). Samples for palaeomagnetic study were drilled at twenty-two sites distributed over the outcrop of the complex (see Fig. 1).

At Rushton to the north west (2°35′ W, 52°40′ N), a poorly-exposed 4-km^2 area of quartz–mica–garnet schist (palaeomagnetic site 31, see Fig. 2) is overlain

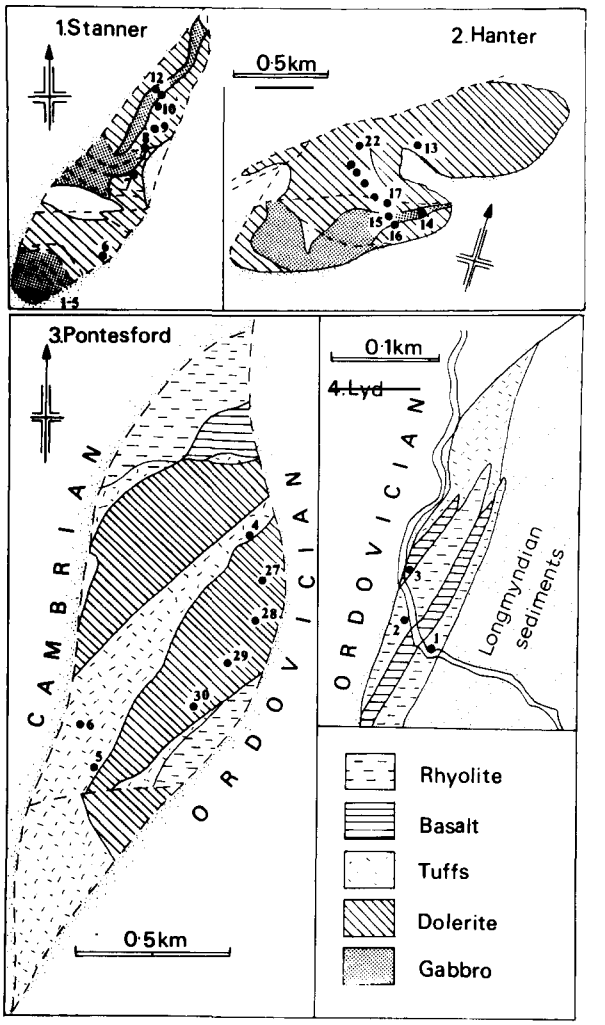


Fig. 1. Geology and sampling localities in Late Precambrian inliers at: 1, Stanner; 2, Hanter (356.9°E, 52.2°N): 3, Pontesford; and 4, Lyd Hole (357.1°E, 52.7°N), Welsh Borderlands.

unconformably by Lower Cambrian quartzites. These (Rushton) schists presumably predate a major non-metamorphic volcanic group comprising the Uriconian episode, which includes thicknesses, locally exceeding 1200 m, of lavas and tuffs (both welded and unwelded) of basaltic, andesitic and dacitic composition; present outcrops were mostly upfaulted close to the major NE–SW trending Church Stretton Fault zone (Fig. 2). This survey covers the Wrekin range and adjoining outcrops (Fig. 2) forming the "Eastern Uriconian" which was folded and intruded

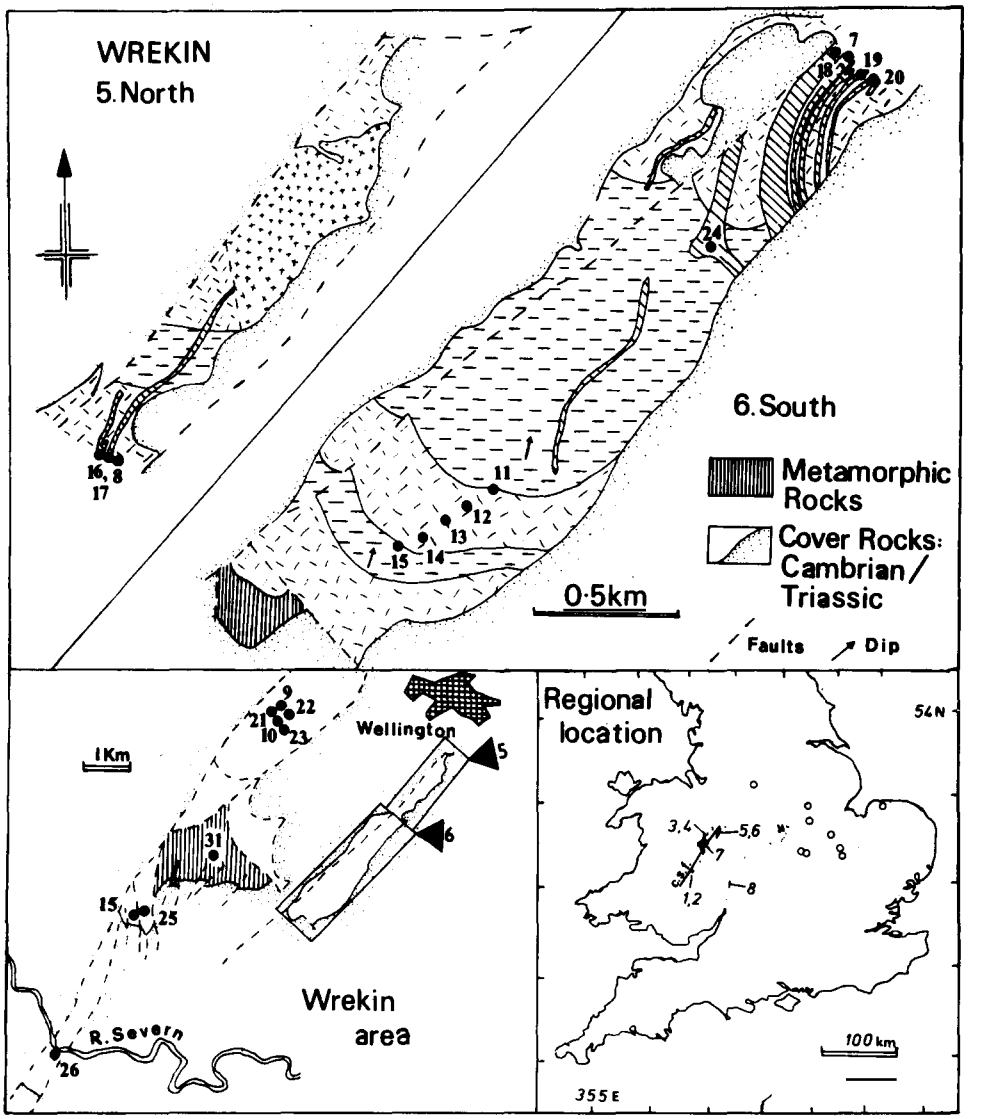


Fig. 2. Geology and sampling sites in the Precambrian inlier of the Wrekin area, Welsh Borderlands. This inlier comprises predominantly Late Precambrian volcanics assigned to the Eastern Uriconian but includes two small outcrops of low-grade metamorphic rocks including the Rushton Schist (site 31). The regional location inset map illustrates the sample localities of Table III in relationship to the distribution of the Precambrian outcrops of the Midland Craton of England and Wales (black) and borehole occurrences open circles): 1, Stanner; 2, Hanter; 3, Pontesford; 4, Lyd; 5 and 6, Wrekin; 7, Longmynd; and 8, Malvernian. c.s.f. = Church Stretton Fault.

by dolerites (Pocock and Whitehead, 1948) prior to deposition of Lower Cambrian quartzites. On the west side of the Church Stretton Fault a few outcrops of comparable volcanics occur and these are grouped together as the "Western Uriconian". An important section occurs at Lyd Hole in Pontesford Brook, where basalt and rhyolite flows (Fig. 1, palaeomagnetic sites 1–3) are overlain with minor unconformity by basal Longmyndian sediments (Pocock et al., 1938), a major group of clastic rocks deposited and folded prior to Cambrian times. Nearby, a fault-bound inlier of volcanics at Pontesford Hill (palaeomagnetic sites 4–6) is cut by dolerite sills (sites 27–29). Bath (1974) determined Rb–Sr whole rock

isochrons regarded as threshold dewatering ages of 529, 529, and 452 m.y. on Longmyndian shales and estimated a deposition age of 600 m.y. by extrapolating initial Sr^{87}/Sr^{86} ratios back to a single overall minimum ratio for the source materials. This estimate would be broadly consistent with K–Ar whole-rock ages from the older Uriconian volcanics of 632 ± 32 m.y. from rhyolite (Wrekin range), 638 ± 81 m.y. for a basalt dyke from the same area, and 677 ± 72 m.y. from a rhyolite flow at Pontesford (Fitch et al., 1969).

Previous palaeomagnetic studies of the Precambrian of England and Wales comprise an early study of Longmyndian sediments by Creer (1957). During a second investigation of this group, Lomax and Briden (1977) found a remanence established to be secondary and postdating the late Precambrian folding. These latter authors also examined Uriconian lavas, mostly from the Caer Caradoc region bordering the Longmyndian outcrop, but were unable to establish any primary magnetic directions from total NRMs or AF-cleaned samples, a result suggesting that thermal demagnetisation should accompany any further study. Since the group mean was similar to the mean derived from Longmyndian sediments, it is probable that a major remagnetisation event affected this area in late Precambrian times, because, although this magnetisation is not significantly different from the present geomagnetic field direction, the remanence of the Longmyndian is definitely ancient, since it predates the magnetism carried by an E–W dyke swarm (Lomax and Briden, 1977) established by studies elsewhere (Piper, 1978) to be Lower Silurian (Llandovery) in age.

3. Palaeomagnetic results

Sampling, measurements and analysis were undertaken according to standard procedures: cores were collected in the field with a motorised drill and oriented by sights on the sun and/or located topographic features; structural dips were measured on tuffaceous units in the field or taken directly from Geological Survey 1: 63,630 maps. All samples were progressively demagnetised in increasing steps of 5 or 10 milliTesla (mT) by 500 Hz alternating fields in a two-axis tumbler, and were measured at each stage with a parasitic magnetometer. Alternating field treatment was continued for all samples until directions of magnetism showed no further systematic or significant changes, and progressive thermal demagnetisation was then commenced, in steps of 150° but reduced to steps of 20 or 30° near the Curie point. This two-stage cleaning treatment has been employed before in the study of Precambrian rocks (see, e.g., Roy and Lapointe, 1976), and may permit the recovery of discrete magnetisations that cannot be recovered by a one-stage cleaning technique. In this study, the thermal demagnetisation was undertaken on one pilot sample from each site with a view to treating the remaining samples from the site at one or two temperature ranges immediately below the Curie point or points, and to recovering any higher-blocking-temperature components present in different oxide phases. No magnetisations with coercivities less than 20 mT are considered in this study.

The gabbros and dolerites from the Stanner and Hanter complex exhibit variable stabilities to AF demagnetisation, but all show an exponential fall in magnetic moments (see, e.g., Fig. 3), and a progressive movement over the first 10–20 mT of treatment which, in the more stable samples, is systematically away from the present Earth's field direction. At higher fields the stabilities are very variable, but movements are not systematic and some samples become too weak to measure (<0.03 A.m^2/kg) after cleaning in fields above 40 mT. Thermal demagnetisation of the stronger samples confirms that the AF technique has adequately recovered the stable NRM and suggests that these moments reside in components with high blocking-temperatures close to the Curie point of magnetite (Fig. 3). This conclusion is supported by thermomagnetic determinations which yield single Curie points in the range 415–565°C (Table I) at all but four sites. The exceptions (sites 5, 11, 13 and 22) show a continuous fall in saturation magnetism on heating in air and convert to a constituent with much higher saturation magnetisation, and on cooling exhibit a Curie point attributable to titanomagnetite; in these instances, the primary remanence carrier is probably pyrrhotite because a significant sulphide content (pyrrhotite and/or pyrite) might oxidise to magnetite on heating to produce the type D curves of Table I and Fig. 4.

Of the twenty sites yielding significant groupings

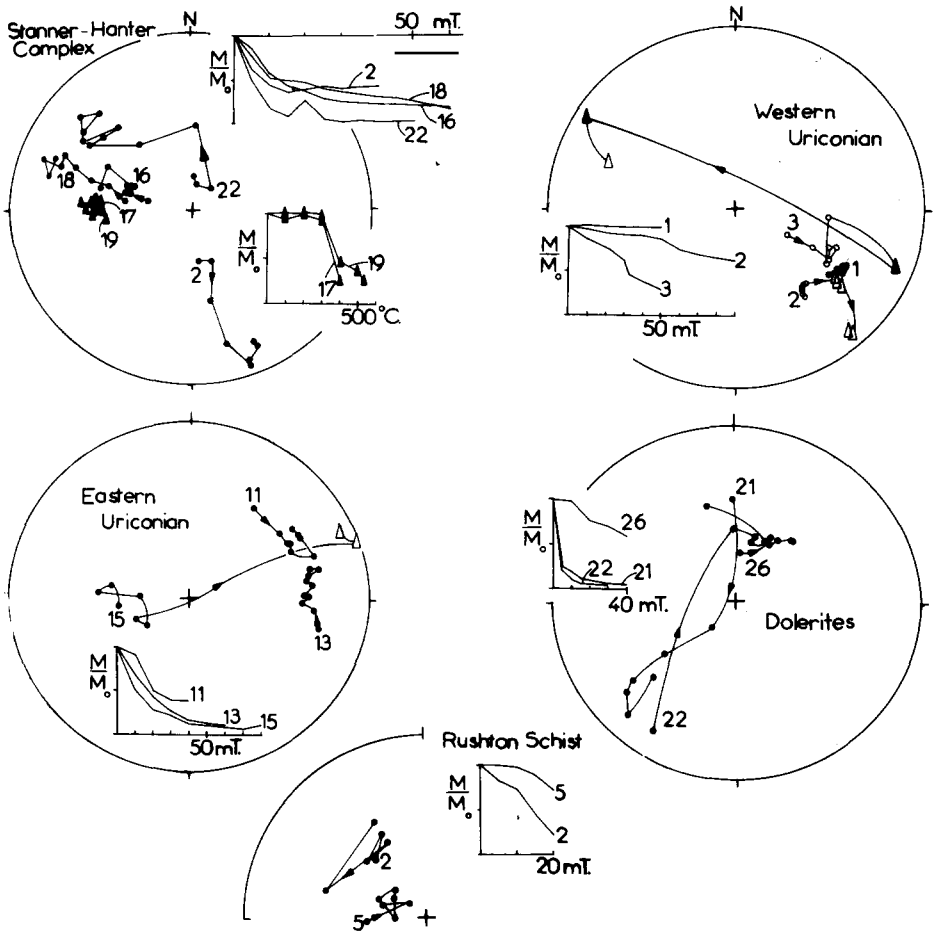


Fig. 3. Behaviour of some typical sample directions of magnetisation with progressive AF (circles) or thermal (triangles) cleaning. Polar stereographic projections, closed symbols are lower-hemisphere and open symbols are upper-hemisphere plots. The decay of NRMs with progressive cleaning is represented as a fraction of the initial moment (M_0).

of directions ($p = 0.05$), 15 have westerly declinations and moderate positive inclinations; they yield a group mean of $D = 282°, I = 51°$ ($\alpha_{95} = 11.4°$) which is unlike any post-Cambrian field direction for Britain. If the gabbros and later dolerites are treated separately, they yield group means of $D = 279°, I = 50°$ (8 sites, $R = 7.4$), and $D = 286°, I = 52°$ (7 sites, $R = 6.4$) which are not significantly different from one another. This implies that the two intrusive episodes are closely linked but the direction of the magnetisation of the dolerites is poorly defined: there is a wide variation of magnetic inclination (55°) in these bodies, which also include a stable site 20 ($D = 89°$, $I = 70°$) not incorporated in the above calculation, and it is possible that the collective group of sites includes transitional field directions. It is thought unlikely that the low grade metamorphic episode, probably closely linked with the dolerite intrusive episode, was sufficiently protracted to achieve complete remagnetisation of titanomagnetites with blocking temperatures in the range 400–565°C (Pullaiah et al., 1975). This is probable both because primary pyroxene is locally preserved and because a few sites are widely removed from the group mean; sites 2 and 13 in close proximity to adjacent sites yield stable titanomagnetite-held remanence with a mean of $D = 171°, I = -7°$, also suggesting some subsidiary field directions are present, although no precise reversals of

TABLE I

Palaeomagnetic results: igneous rocks of Stanner and Hanter Hills, Radnorshire, Welsh Borderlands (356.9°E, 52.5°N)

Site no.	Rock type	N	M_0 ($\times 10^{-5}$ A.m^2/kg)	D (°)	I (°)	R	k	α_{95} (°)	Curie point (°C, ±10°C)	Thermo-magnetic curve type
1 *	G	4	2.42–0.04	43	66	3.85	20	21	550	C
2 *	G	6	0.36–0.07	167	−14	5.86	37	11	555	C
3	G	5	0.08–0.02	268	41	4.72	14	21	460	C
4 *	G	3	3.92–0.42	309	−15	2.93	26	24	450	C
5	G	6	0.07–0.01	252	47	5.78	23	14	−/415	D
6	D	5	1.36–0.87	277	58	4.50	8	29	550	A
8	D	5	0.11–0.04	275	81	4.83	23	16	550	C
10	D	7	0.06–0.02	271	39	5.89	5	28	545	C
11	G	6	0.09–0.01	304	33	4.76	4	38	−/550	D
12	G	6	0.05–0.01	212	57	5.89	38	11	485/535	C
13 *	D	4	3.14–0.07	174	0	3.64	8	34	−/565	D
14	G	5	0.24–0.01	320	48	4.52	8	28	535	C
15	G	5	0.02–0.01	290	30	3.52	3	58	545	C
16	G	5	1.32–0.31	279	56	4.68	13	22	550	C
17	G	4	2.60–0.11	275	56	3.42	5	45	545	C
18	D	8	0.87–0.01	265	39	7.12	8	21	555	C
19	D	4	1.00–0.35	259	51	3.64	8	33	490	A
20 *	D	5	0.75–0.23	89	70	4.87	31	14	470	A
21	D	8	1.81–0.39	331	50	7.72	25	11	525	A
22	D	5	0.84–0.07	313	26	4.58	10	26	−/415	D
Mean of 15 sites				281.7	50.8	13.85	12	11.4		

Palaeomagnetic pole: 279.1°E, 31.4°N (dp = 10.4°, dm = 15.4°)

G = gabbro, D = dolerite, * = sites excluded from the group mean calculation. D and I are the declination and inclination of the mean direction of magnetisation of N samples with R equal to the resultant vector, k [= $(N-1)/(N-R)$] is the Fisher precision estimate, and dp and dm are the radii of the 95% confidence oval about the mean pole, in the colatitude direction and at right angles to it, respectively. Heating and cooling Curie points are listed respectively where they differ. Thermomagnetic curves are classed as follows: A, heating and cooling curves similar; B, cooling curve below heating curve; C, cooling curve above heating curve; D, very large increase in saturation magnetisation on cooling.

magnetisation have been recorded in this sample.

Specimens of Rushton Schist have weak magnetic moments moderately stable to AF treatment (Fig. 3) and seven samples group after cleaning to yield a mean of $D = 304°, I = 39°$. When this is corrected for dip of the overlying Cambrian sediments (30° to the south) the mean is moved to $D = 274°, I = 51°$ (site 31, Table II). This is again unlike any Ordovician or late Palaeomagnetic direction from Britain but is similar to the mean direction for the Stanner–Hanter complex. Unfortunately, the paucity of in situ outcrop of the Schist does not allow this result to be defined with greater precision, and the samples could not be thermally treated because they undergo drastic alteration on heating (see Fig. 4): a sharp drop in saturation remanence continues to temperatures of about 250°C and is also indicative of pyrrhotite as a remanence carrier; this mineral probably oxidises to magnetite ($T_c \sim 560°C$) to yield the very large increase in saturation magnetisation observed on cooling.

All the remaining sites of this study are derived from volcanic rocks included in the Uriconian Group (Dunning, 1975), or are dolerite sills and dykes cutting these volcanics; a summary of palaeomagnetic results is listed in Table II.

Rhyolite sites 1 and 2 from the Lyd Hole section possess a high coercivity magnetic remanence which is very stable to AF treatment (see Fig. 3); with progressive thermal treatment the direction of magnetisation

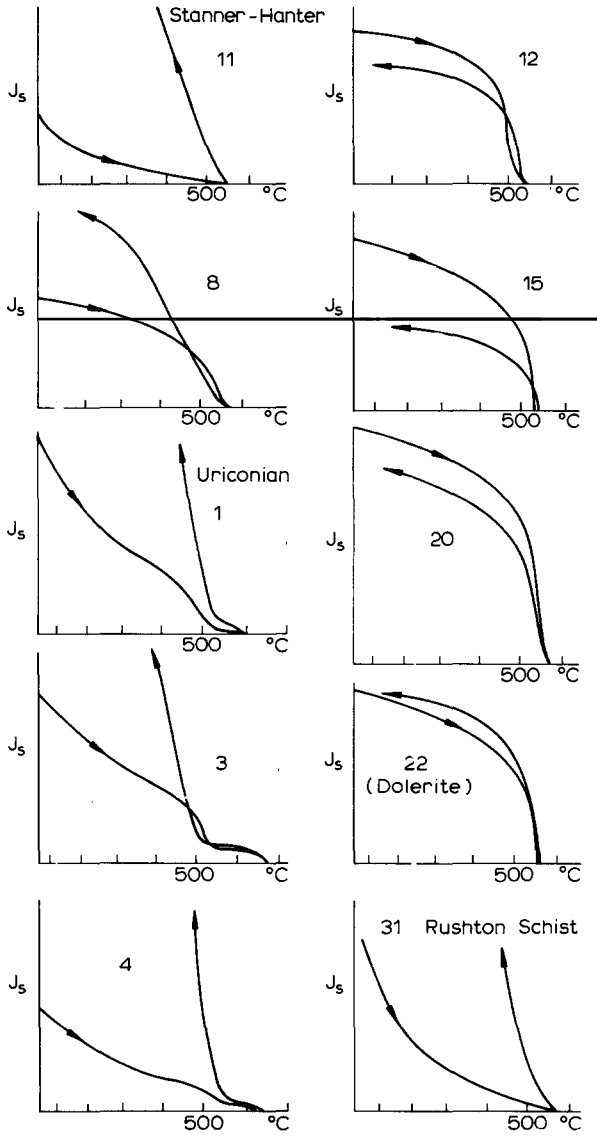


Fig. 4. Some representative thermomagnetic determinations (saturation magnetisation, J_s versus temperature in degrees Celsius) from the palaeomagnetic collections; the J_s axis is arbitrary.

of site 2 does not change, but the inclination of site 1 decreases to more nearly approach that of site 2, and demagnetisation data for this site at 630°C are listed in Table II. Basalt site 3 is less stable, and yields a mean of $D = 118°, I = 39°$ after AF treatment, but since this direction prior to tilt correction is similar to the secondary magnetisation held by the overlying Longmyndian sediments (Lomax and Briden, 1977), it cannot be regarded as a primary deuteric magnetisation. However, with thermal treatment the sample directions of magnetism show a progressive movement, and above 600°C all samples acquire a closely antiparallel direction (Fig. 3) to those of sites 1 and 2 with about 1% of initial remanence remaining. The mean after cleaning at 630°C is listed in Table II; since this is antiparallel to those of other sites in this area, it is probably an ancient chemical remanent magnetisation (CRM) acquired in a reversed field. The complex behaviour of these sites is endorsed by thermomagnetic determinations (Fig. 4, Table II); in addition to a Curie point attributable to magnetite, they also show a second, higher Curie point attributable to hematite (590–670°C). Other lava sites (4 and 5) from Pontesford show more simple directional behaviour with cleaning treatment; the second thermal stage of treatment yields no magnetisations not isolated by the AF treatment and there is no significant difference between the site mean directions computed from sample directions after AF and after thermal cleaning. Site 6 in a tuffaceous unit was too weakly magnetised to be thermally treated but moved, after AF cleaning, to a mean direction close to those of the other sites. The overall mean, assuming the site mean direction for site 3 to be a reversed field direction, is listed in Table II after tilt corrections of individual sites for dips of 57–61° to the SE.

Most lavas belonging to the eastern Uriconian outcrop have single, high-temperature Curie points somewhat below that of pure magnetite (Fig. 4 and Table II). Four sites (12–15, Table II) showed some change on heating with a slightly higher Curie point after the heating process; in these cases oxidation has probably taken place to produce a host titanomagnetite closer to pure magnetite in composition; site 9 has a residual saturation moment due to hematite. In most cases demagnetisation behaviour was correspondingly simple; samples show progressive movement to stable end points with AF treatment, and the mean directions computed after AF and after thermal demagnetisation are comparable. The exceptions are the poorly-stable site 7 (AF data listed in Table II) which showed no systematic behaviour with thermal treatment, and the (Charlton Hill) andesite site 15 which showed a large change in site mean direction above 500°C; since the mean of this latter site evaluated by

TABLE II

Palaeomagnetic results: Uriconian lavas and intrusions, and Rushton schist, Wrekin and Pontesford inliers, Welsh Borderlands (357.1–357.5°E, 52.7°N)

Site No.	M_0 ($\times 10^{-5}$ A.m^2/kg)	N (samples)	R	k	Direction of magnetisation (°) before and after tilt correction				Curie point(s) (°C, ±10°C)	Thermo-magnetic curve type
					Before		After			
					D	I	D	I		
(i)	*Western Uriconian Volcanics*									
1	0.32–0.02	6	5.53	11	137	51	137	−8	535, 590	D
2	0.80–0.14	8	7.69	22	130	21	129	−36	525, 610	A
3	0.28–0.03	7	5.98	6	323	−52	315	5	530, 670	D
4	1.19–0.14	6	5.95	105	139	25	139	−35	550, 630	D
5	0.38–0.10	6	5.88	40	139	2	133	−49	545	D
6	0.34–0.08	6	5.69	17	133	29	139	−14	–	–
(ii)	*Eastern Uriconian Volcanics*									
7	0.55–0.28	5	4.91	45	85	61	89	2	550	A
8	0.13–0.01	7	6.78	28	130	83	84	26	555	B
9	3.13–0.12	4	3.86	22	65	−13	66	27	555, 665	C
10	2.26–0.76	4	3.81	16	93	−2	102	34	560	B
11	1.55–0.08	6	5.92	65	351	72	64	30	525	C
12	0.23–0.04	2	1.95	22	57	48	75	−7	505/530	B
13	2.16–0.33	6	5.93	71	51	69	77	14	510/530	B
14	0.81–0.11	4	3.98	179	56	82	83	27	535/545	D
15	21.70–17.40	6	4.56	3	100	−43	68	−7	520/535	B
(iii)	*Post-Uriconian Dolerites*									
16	6.05–0.31	8	7.58	17	77	78	123	53	550	A
17 *	1.65–0.40	5	4.95	73	332	−54	328	−25	540	A
18	5.60–2.04	7	6.78	28	17	50	59	57	560	A
19	0.23–0.09	3	2.99	1423	17	47	55	55	530	B
20	4.08–1.16	6	5.66	15	43	38	43	38	560	B
21 **	4.79–0.35	6	5.64	14	224	17	–	–	550	A
22	21.90–1.80	4	3.92	36	20	68	–	–	550	A
23 **	1.55–0.13	4	3.80	15	214	21	–	–	555	A
24	4.04–0.06	8	7.60	18	66	64	102	46	545/570	C
25 **	1.06–0.08	6	5.86	36	231	52	213	43	530, 625	D
26	2.44–0.09	7	6.70	20	23	46	–	–	520	A
27	5.75–1.94	7	6.86	44	23	25	73	36	525	B
28	1.87–0.78	7	6.79	29	345	29	86	59	535	B
29	4.34–1.54	7	6.50	12	354	40	102	58	525	B
30	20.70–3.68	6	5.38	8	27	41	92	34	535	B
(iv)	*Rushton Schist*									
31	0.03–0.02	7	6.52	12	301	39	274	51	~250, 530	D

Group		N (sites)	Group mean directions					Palaeomagnetic poles	
			R	k	D	I	α_{95}	°E	°N
(i)	*Western Uriconian Volcanics*	6	5.75	20	135.8	−24.5	15.3	55.3 (dp = 8.8°, dm = 16.4°)	−36.9
(ii)	*Eastern Uriconian Volcanics*	9	8.54	17	78.3	16.8	12.8	91.5 (dp = 6.8°, dm = 13.2°)	13.9
(iii)	*Post Uriconian Dolerites* **	3	2.92	26	218.5	27.2	24.6	318.9 (dp = 14.6°, dm = 26.8°)	−15.2
(iv)	*Post Uriconian Dolerites, main group*	11	10.22	13	71.9	54.2	13.2	77.1 (dp = 13.0°, dm = 18.6°)	37.4
(v)	*Rushton Schist*	7 samples	–	–	–	–	17.8	285	26

Symbols are the same as those used in Table I.

* Sites excluded from group mean. ** Three sites with magnetisations unrelated to main group (see text).

AF treatment is close to the present field direction prior to dip correction, the thermally-cleaned mean is accepted (Table II). Site mean directions after thermal cleaning are also listed for sites 6 and 7 where this method yielded a significant improvement in within-site grouping. The Uriconian lavas have been folded at least twice, firstly in Late Precambrian times and secondly in Lower Palaeozoic (Caledonian) times. In the case of sites 7, 8 and 11–15 it is possible to correct firstly for the tilt of the unconformably-overlying Lower Cambrian sediments and secondly for the residual tilt of the lavas acquired in Precambrian times before the Palaeozoic tilt was imposed; sites 9 and 10, however, are remote from any cover rocks and the best that can be achieved here is to correct them for the dip measured at outcrop. Tectonic orientations at the different sampling localities are quite different and the correction for tilt yields a striking improvement in precision factor (k) from 3 to 17 and a ratio of k (after) : k (before) which is significant at the 95% confidence level; this test strongly suggests that the AF/thermally-derived mean directions are Precambrian in age. The two-stage tilt correction is quite critical to this improvement in grouping of the Eastern Uriconian, but cannot explain the difference between the mean directions of the Eastern and Western Uriconian, because in the latter case the strikes of the Precambrian and unconformably-overlying Ordovician are nearly identical and a combined single tilt correction is appropriate (see Fig. 5). This difference is thus taken to reflect an unknown age difference between the two. Although none of the present separate outcrops of the Uriconian Group can be correlated with one another (Dunning, 1975), the petrologic similarities between these outcrops suggest that this difference is unlikely to be very great.

With the exception of site 25, the dolerite intrusions cutting the Uriconian lavas all have simple thermomagnetic spectra (see Fig. 4 and Table II) indicating a single magnetic component close to pure magnetite in composition. The mean values (after AF cleaning throughout) are listed in Table II; thermal cleaning appeared to promote chemical changes and was only partially successful for this collection; since thermomagnetic determinations (Table II) only located hematite at one site, this is probably not an important disadvantage. In the Wrekin area, the dolerites are seen to cut folded Uriconian lavas

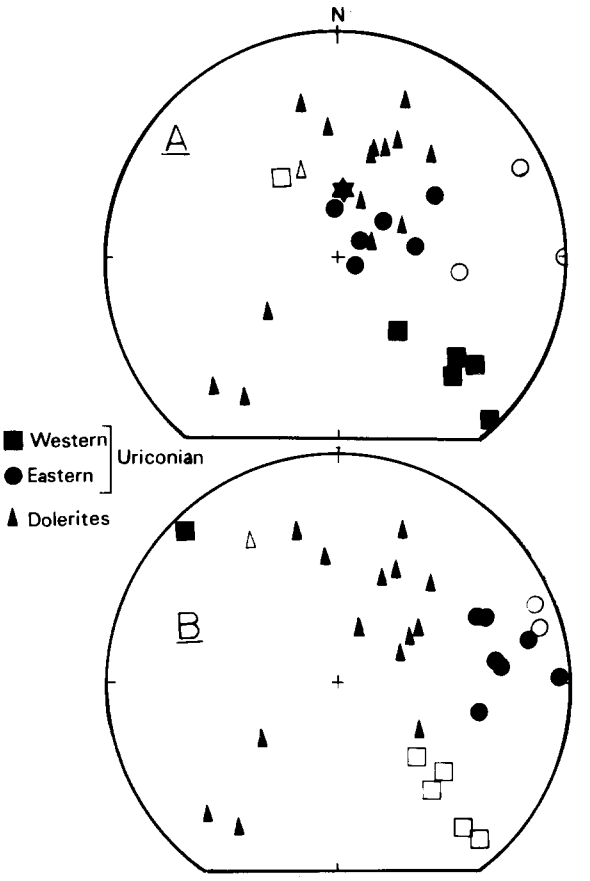


Fig. 5. Site mean directions of magnetisation after AF and thermal cleaning (A) before tilt correction, and (B) after tilt correction. Solid symbols are lower-hemisphere plots and open symbols are upper-hemisphere plots; the star is the direction of the present mean axial dipole field. Stereographic projection.

and tuffs and sites 16–20, 24 and 25 are accordingly corrected only for tilt of the overlying basal Cambrian rocks; no post-intrusion tilt is evident for dykes 21–23 and 26 for which in situ directions are accepted. Site 17 is anomalous both before and after correction, while site 25 from the southern part of the inlier yields a mean close to those of sites 21 and 23 from the northern part, suggesting that an intermediate field direction is represented, or that more than one episode of dolerite intrusion is present in this area (the mean of these three sites is listed in Table II). In view of these intermediate directions and the restricted outcrop of the dolerites, no effective

fold test can be applied to these data.

The Pontesford dolerites occur as sill-like bodies invading Uriconian volcanics, and, since they do not intrude the overlying Longmyndian sediments, they are probably older and were hence involved in the Late Precambrian folding of the sediments. The removal of secondary magnetisations in sites 27–30 by AF treatment yields mean directions which are also very stable to thermal demagnetisation (see Fig. 3) although the overall grouping improves slightly with this latter technique. The mean directions after cleaning at 530°C are listed in Table II. When tilt corrected, these directions group with dolerites from the Wrekin area although they are quite different prior to this correction. The combined mean for all of the dolerites after AF and thermal cleaning is then $D = 72°, I = 54°$ ($\alpha_{95} = 13$).

Although the field and laboratory tests have enabled directions of magnetisation of undoubted primary origin to be recovered on this study, the qualities of the group means are ultimately restricted by the tectonic information relating to the history of deformation. Since some of the sampled units have been folded at least twice and occur within fault-bounded blocks, this is necessarily imprecise and must contribute to increasing the overall scatter of magnetic directions of each group. However, we can be encouraged, firstly by the recovery of directions of magnetisation quite unlike any known from post-Cambrian times in general and times of Caledonian folding (~500–380 m.y.) in particular, secondly by the agreement of results from the dolerite intrusions into the Eastern and Western Uriconian after correction for Late Precambrian and later folding, and finally by a positive fold test from the Eastern Uriconian volcanics. The group mean directions of magnetisation might in future be more precisely defined by extending the study to the remaining Uriconian outcrops. However, the study of Lomax and Briden (1977) suggests that these outcrops have a prominent secondary magnetisation, and, since they are all fault-bound inliers and have steep tilts, the tectonic problem will remain. Anisotropy of magnetic susceptibility has not been determined in this study, but it is a visible property only in the Rushton Schists; the close agreement of directions of magnetisation between this rock unit and metamorphic rocks of the same region, specifically the Stanner–Hanter complex and the Malvernian metamorphic rocks (Brooks, 1963; Lomax and Briden, 1977), suggests that this property has not greatly influenced the mean direction of magnetisation.

4. The time sequence of palaeomagnetic poles

The time sequence of palaeomagnetic poles from a single cratonic unit defines an a.p.w. path, and such a path can now be established for cratonic Britain [the Midland Craton of Watson (1975)] using the results of this study, together with some previously-published data from the same crystal block and spanning the comparable interval of Late Precambrian–Lower Cambrian times. These poles are listed in Table III.

The Malvernian rocks have been attributed both to Middle or Lower Proterozoic times (Baker, 1971; Shackleton, 1969) and to late Proterozoic times (Wright, 1969; Thorpe, 1972), but as noted by Dunning (1975) the metamorphic grade of the basement rocks within the Midland Craton is appreciably lower than the (presumably older) basement rocks of the Baltic and Laurentian Shields of Middle or Lower Proterozoic age (Laxfordian, Svecofennian and Hudsonian). Also, no radiometric determination older than 700 m.y. has yet been made on igneous or metamorphic rocks of the Midland Craton (Fitch et al., 1969; P.J. Patchett, personal communication) and it is possible that the entire exposed area of the region is Late Precambrian in age. Poles 1–3 from the metamorphic rocks of this Craton lie in close proximity to one another, although the sampled areas are up to 60 km apart. The age relationship of the two poles from the Eastern and Western Uriconian (entries 4 and 5) is unknown but they are listed in the order involving the least polar movement between them and the pole for the later post-Uriconian dolerites, and three poles then form a sequence following on directly from the poles 1–3 from low-grade basement rocks (Fig. 6). The Uriconian poles also lie in the same vicinity as two poles (poles 9 and 10 in Table III) from volcanic rocks of North Wales, dated by reference to the faunal content of associated sediments as Lower Cambrian (~570 m.y.) in age. However, a.p.w. movements in the interval between the eruption of the Uriconian Group and Lower Cambrian times may have been more complex than suggested by the distributions of

TABLE III

Precambrian palaeomagnetic poles for cratonic Britain south of the Caledonian Suture

Rock unit	Age (m.y.)	Pole position	dp°/dm°	Sites/ samples	References * and comments
(i) Basement rocks of Upper (?) Proterozoic age					
1. Rushton Schist	–	285°E, 26°N	–/–	1/7	[2]
2. Stanner–Hanter Complex	–	279°E, 31°N	10/15	15/84	[2]
3. Malvernian Diorites	–	252°E, 13°N	11/23	4/26	[3]; tilt-corrected in situ direction is statistically identical to present field
(ii) Late Precambrian–Lower Cambrian rocks (largely unmetamosphosed)					
4. Western Uriconian Volcanics	677 ± 72 ** [1]	55°E, 37°S	9/16	6/39	[2]
5. Eastern Uriconian Volcanics	632 ± 32 ** [1]	92°E, 14°N	7/13	9/44	[2]; positive fold test
6. Post-Uriconian Dolerite Intrusions	638 ± 81 [1]	77°E, 37°N	13/19	11/70	[2]
Interval of Late Precambrian times represented by deposition of clastic sediments (Longmyndian), folding of Uriconian and Longmyndian, and remagnetisation of entries 7, and possibly 8, below					
7. Longmyndian Group secondary magnetisation	600 ***	29°E, 60°N	11/11	69/464	[3]; negative fold test but magnetisation definitely pre-Lower Silurian [5]
8. Uriconian Group secondary magnetisation	–	41°E, 80°N	23/24	13/86	[3]; negative fold test, most sites appear to carry secondary magnetisation
9. Gwna Group Pillow Lavas, Anglesey, Wales	Lower Cambrian	53°E, 17°S	14/28	4/19	[4]; one reversed site but no satisfactory field test
10. Arvonian Volcanics, North Wales	Lower Cambrian	33°E, 51°S	–/–	6/36	[3]; poorly-defined mean direction (α_{95} = 38°) but approaches that of rock unit 9 after separate and quite different tilt corrections

* Reference abbreviations: [1], Fitch et al. (1969); [2], this work; [3], Lomax and Briden (1977); [4], Piper (1976b); [5], Piper (1978).

** K–Ar whole-rock determinations.

*** Estimated from Rb–Sr data.

Entries 3, 7, 8 and 10 include sites at total NRM level.

poles 1–6 and 9 and 10 because the sediments of the Longmyndian Group overlying the Uriconian lavas possess a secondary magnetisation [see pole 7, Table III, taken from Lomax and Briden (1977)] which is definitely pre-Silurian in age (Piper, 1978) and was thus imposed prior to the widespread remagnetisation found in the Anglo-Welsh cuvette by Chamalaun and Creer (1964). It probably dates from the Late Precambrian folding (~600 m.y.) of the Longmyndian Group because a similar secondary magnetisation (pole 8, Table III) is present in Uriconian rocks adjoining the Longmyndian outcrop. Extensive demagnetisation studies (R.L. Smith, personal communication) suggest that the magnetisations yielding poles 7 and 8 are composite and includes a present-field component. With comprehensive thermal demagnetisation sample directions migrate away from the present field axis and yield a palaeopole nearer that

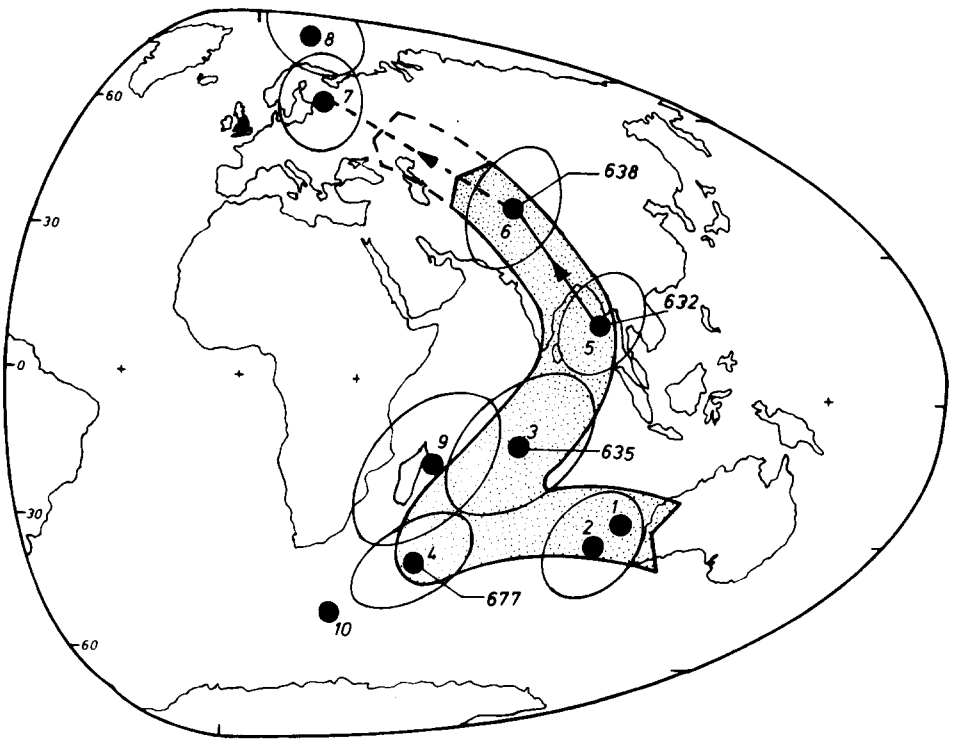


Fig. 6. Late Precambrian–Lower Cambrian palaeomagnetic poles for the Midland Craton listed in Table III; 95%-confidence ovals are given, together with minimum ages from K–Ar whole-rock determinations or Rb–Sr data (pole 3).

of the post-Uriconian dolerites (pole 6); this magnetisation still, however, postdates the Late Precambrian folding of the Longmyndian Group. In the absence of palaeomagnetic poles from primary magnetisations spanning the interval between the Uriconian volcanism and Lower Cambrian times, the form of the a.p.w. path in this interval remains largely speculative.

5. Discussion

Poles 1–6, 9 and 10 (in Table III and Fig. 6) are displaced by 30–90° from the Ordovician south pole for Britain (Briden et al., 1973) and are hence probably south poles. A large a.p.w. movement contemporaneous with these poles is identified from Gondwanaland data (McElhinny et al., 1974; Piper, 1976a) but there are currently still insufficient data for the Midland Craton to correlate the two curves in detail. In Fig. 7 the poles from the Midland Craton are compared with Late Precambrian palaeomagnetic data from the Baltic Shield, and it is seen that basement poles 1–3 from the Midland Craton define a field distinct from poles for rock units within the Sveconorwegian mobile belt (~1000–700 m.y.) with which they might be contemporaneous. However, we also observe that these poles lie close to the oldest post-metamorphic Uriconian poles, suggesting that they might not be appreciably older than 670 m.y., and consistent with Rb–Sr data relating in part to pole 3 and reported by Lambert [in Fitch et al. (1969)].

Upper Silurian to Lower Devonian palaeomagnetic results from Norway define a geomagnetic field axis near $D = 210°, I = 10°$ (Storetvedt, 1967) which differs from the contemporaneous field axis for Britain of $D = 220°, I = 55°$ (Briden et al., 1973), and the two regions appear to have joined along a suture zone in the present North Sea region in post Lower Devonian times and prior to Upper Devonian times (Morris, 1976). If we superimpose the Upper Silurian–Lower Devonian poles from Britain and Norway it does not bring the Upper Proterozoic poles into coincidence

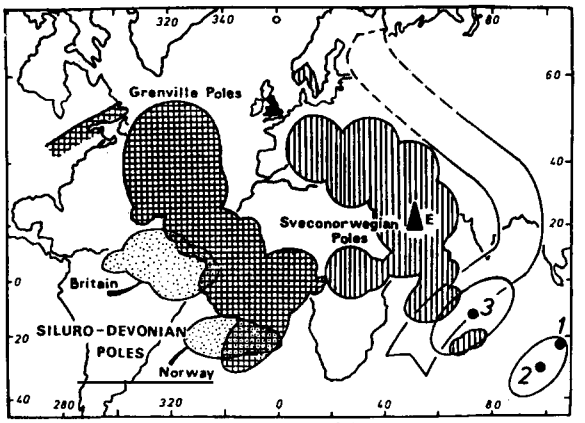


Fig. 7. The field occupied by palaeomagnetic poles from the Sveconorwegian Province (~1100–800 m.y.) of the Baltic Shield (Patchett and Bylund, 1977) compared with Upper Proterozoic poles from the Midland Craton (poles 1–3 in Table III). Also shown is the apparent polar wander swathe for this craton in late Precambrian times defined by poles 4–8, together with the pole for the Egersund Dolerite (663 m.y., triangle, pole E). The field of palaeomagnetic poles from the Grenville Province of North America (contemporaneous with the Sveconorwegian Province) is also illustrated on present-day coordinates, together with the fields of British and Norwegian Silurian–Lower Devonian poles (see Section 5).

(Fig. 7). Indeed it moves all of these poles further apart and, in particular, the Egersund Dolerite pole (triangle in Fig. 7, ~663 m.y., and the only current Scandinavian result approximately contemporaneous with poles 3–6 of Table III) is removed further from the Uriconian and related poles. It is concluded that relative movements between the Midland Craton and Baltic Shield do not simply involve a convergence in late Lower/Middle Devonian times along a suture near the axis of the present North Sea, but include unspecified Late Precambrian/Lower Palaeozoic motions. More work will be necessary to define these.

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